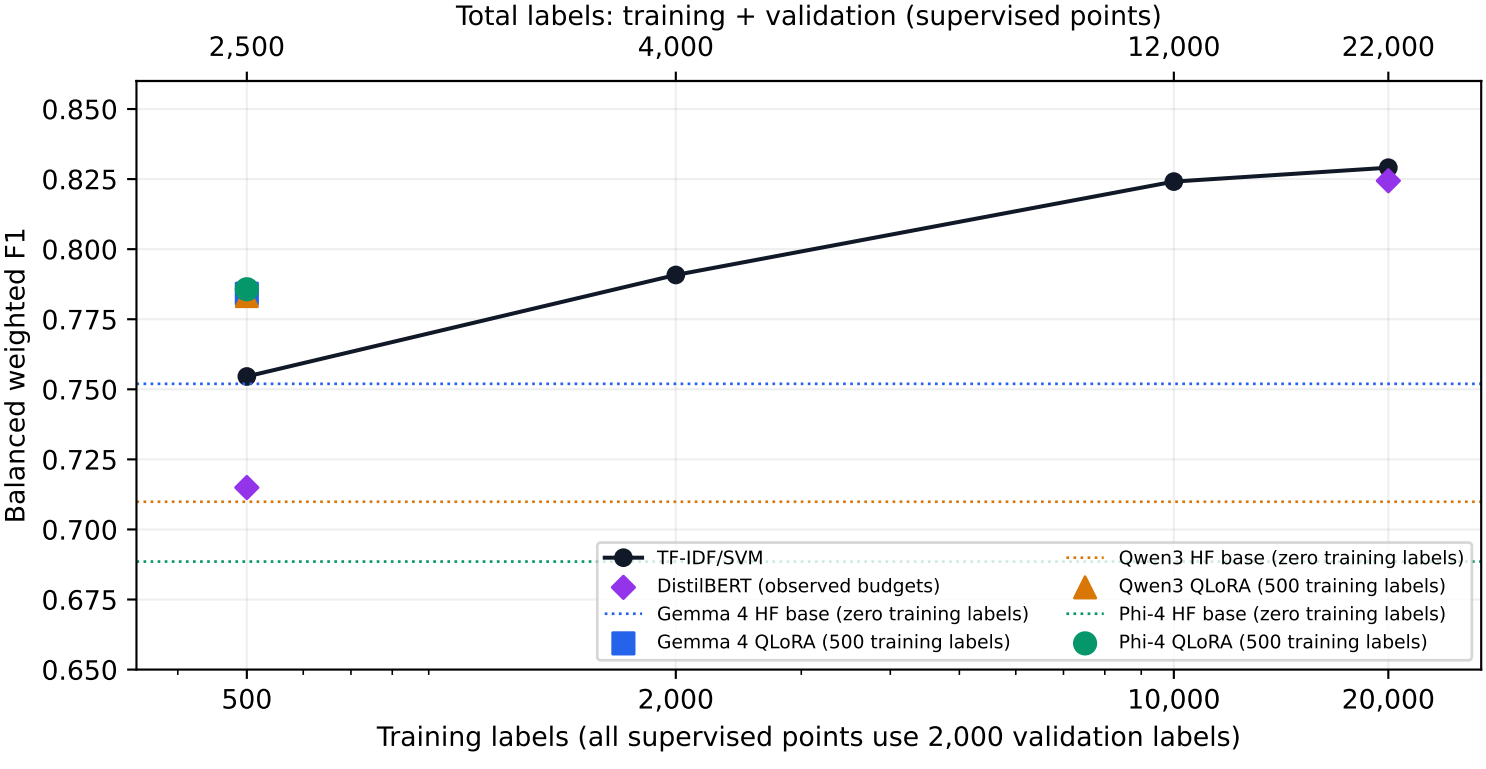
Total labels: training + validation (supervised points)
2,500
4,000
12,000
22,000
Balanced weighted F1
0.850
0.825
0.800
0.775
0.750
0.725
0.700
0.675
0.650
500
2,000
10,000
20,000
Training labels (all supervised points use 2,000 validation labels)
TF-IDF/SVM
DistilBERT (observed budgets)
Gemma 4 HF base (zero training labels)
Gemma 4 QLoRA (500 training labels)
Qwen3 HF base (zero training labels)
Qwen3 QLoRA (500 training labels)
Phi-4 HF base (zero training labels)
Phi-4 QLoRA (500 training labels)